



College of Computing and Informatics
Department of Computer Science

Course Syllabus

Course Title **Networking Fundamentals**

Course Code
1501250

Spring Semester 2024/2025

Course Syllabus

Course Title	Networking Fundamentals										
Course Code	1501250										
Credit Hours	(3 - 0: 3)				Contact Hours			(3 - 0: 3)			
Course Category	☐	Univ. Req.		☐	Col. Req.		☒	Prog. Req.		Others	
Course Type	☒		Compulsory						Elective		
Pre-requisite(s)	1501214 Programming with Data Structures or equivalent										
Co-requisite(s)	None										
Semester	Spring				Year		2024/2025				
Instructor Name	Dr. Bouziane Brik										
Office Location	M5-207										
Tel. No.											
Email	bbrik@sharjah.ac.ae										
Lecture Times											
Office Hours	TBA on the Blackboard										

Course Description:

Foundation knowledge for computer networks, specifically targeting TCP/IP protocol stack. Topics include basic network design, layered communications models (OSI and TCP/IP), the Internet current architecture, application layer protocols (such as HTTP, FTP, SMTP, DNS), unreliable (UDP) and reliable (TCP) data delivery transport layer protocols, internetworking, IP addressing, network layer protocols (such as IP, DHCP, NAT), basic routing algorithms, data link layer protocols (such as ARP, Ethernet), MAC address, medium access protocols (such as channel partitioning, random access, and taking-turns protocols), error detection and correction techniques.

Course Objectives/Goals:

The goals of this course are to:

- Understand basic principles of computer networking.
- Be familiar with the main technologies used in computer networks.

- Comprehend internetworking principles and how internet protocols, routing and applications operate.
- Apply the skills learned to design network applications by making use of tools and network protocols.

Course Learning Outcomes:

By the end of successful completion of this course, the students will be able to:

1. Explain the purpose of network layered models and the function and terminology related to each layer of these models.
2. Analyze the features and operations of various protocols of application, transport, and network layer.
3. Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput.
4. Apply routing algorithms to find shortest paths for network-layer packet delivery.
5. Comprehend the services provided by the data link layer, such as error handling and medium access control.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Analyze the features and operations of various protocols of application, transport, and network layer. - Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput.
	Apply	- Apply routing algorithms to find shortest paths for network-layer packet delivery.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	-
	Implement	-
	Evaluate	- Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput.
Communicate effectively in a variety of professional		-

Computer Science Program SLOs		Course SLOs
contexts.		
Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Apply computer science theory and software development fundamentals to produce computing-based solutions.	Theory	- Explain the purpose of network layered models and the function and terminology related to each layer of these models. - Analyze the features and operations of various protocols of application, transport, and network layer. - Apply routing algorithms to find shortest paths for network-layer packet delivery.
	Development	- Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput. - Comprehend the services provided by the data link layer, such as error handling and medium access control.

IT Multimedia Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Analyze the features and operations of various protocols of application, transport, and network layer. - Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput.
	Apply	- Apply routing algorithms to find shortest paths for network-layer packet delivery.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in	Design	-
	Implement	-

IT Multimedia Program SLOs		Course SLOs
the context of the program's discipline.	Evaluate	- Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput.
Communicate effectively in a variety of professional contexts.		-
Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.	Identify / Analyze	- Explain the purpose of network layered models and the function and terminology related to each layer of these models. - Analyze the features and operations of various protocols of application, transport, and network layer.
	Development	- Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput. - Comprehend the services provided by the data link layer, such as error handling and medium access control.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	Course SLO
1	Introduction: data communication its components and criteria, network topologies, network types, standardization, network services and protocol. Network Models: introduction to TCP/IP and OSI layered models, protocols and service models at each layer, encapsulation.	Ch 1	1
2	The Internet: network edge, access networks, physical media, and the network core, packet and circuit switching, types and causes of network delay and data loss in packet switched networks.	Ch 1	1, 3

3	Application Layer: principles and architecture of network applications, inter-processes communication, transport layer services provided to application layer, application layer protocols: HTTP: persistent and non-persistent connections, http message format, cookies, web caching.	Ch 2	1, 2
4	FTP: file transfer, ftp commands and replies, comparison with http. SMTP: mail message format, mail access protocols, comparison with http. DNS: services provided by DNS, how it works, DNS records and messages.	Ch 2	1, 2
5	Transport Layer: protocols and services provided by transport layer, transport layer interaction, multiplexing and demultiplexing, transport layer protocols:	Ch 3	2
6	User Datagram Protocol (UDP): UDP working, segment structure, checksum.	Ch 3	2
7	Transmission Control Protocol (TCP): TCP service model for Internet, TCP segment structure, connection, flow control, error control.	Ch 3	2, 3
8	Midterm Exam		
9	Network Layer: network layer services, virtual circuit and datagram networks, routers' dissection, network layer protocols:	Ch 4	2
10	Internet Protocol version 4 (IPv4): forwarding and addressing in the Internet, IP datagram format, IP fragmentation and reassembly, IPv4 addressing.	Ch 4	2
11	Dynamic Host Configuration Protocol (DHCP): motivation and usage Network Address Translation (NAT): motivation and usage	Ch 4	2
12	Routing Algorithms: Link-State (LS) and Distance-Vector (DV) routing algorithm, comparison between the two.	Ch 4	4
13	Data Link Layer: service provided by the link layer, location of its implementation. Error detection and correction techniques: parity checks, checksum, Cyclic Redundancy Check (CRC), comparison of checksum and CRC.	Ch 5	5

14	Address Resolution Protocol (ARP): motivation and working, MAC addresses, MAC vs. IP, ARP vs. DNS.	Ch 5	5
15	Multiple Access Protocols: channel partitioning protocols, random access protocols, and taking-turns protocols.	Ch 5	5
16	Final Exam		
	Final Exam		

* In the comments, you can add the relevant chapter or notes, etc.

**Subject to time and space availability

Scheduling of laboratory and other non-lecture sessions, including online sessions, as appropriate (if applicable)

Week	Topic	Comments
	N/A	

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date*	Comments
Homework #1	Week #3	
Homework #2	Week #6	
Homework #3	Week #9	

*The dates are tentative.

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Homeworks	Week #3, 6, 9	15
Quizzes/itbls	Every 2-3 weeks	15
Mid-Term Exam	Week #9	25
Final Exam	Week #16	45
Total		100

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS Competencies
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Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS Competencies
Explain the purpose of network layered models and the function and terminology related to each layer of these models.	- Lectures	- Exams - Assignments - Quizzes/itbls	60% of student achieve 60% of the CLO	Digital literacy
Analyze the features and operations of various protocols of application, transport, and network layer.	- Lectures	- Exams - Assignments - Quizzes/itbls	60% of student achieve 60% of the CLO	Critical and analytical thinking
Evaluate the performance of communication links or network (as a whole) using various metrics, such as latency, bandwidth and throughput.	- Lectures	- Exams - Quizzes/itbls	60% of student achieve 60% of the CLO	Critical and analytical thinking
Apply routing algorithms to find shortest paths for network-layer packet delivery.	- Lectures	- Exams - Assignments	60% of student achieve 60% of the CLO	Problem solving
Comprehend the services provided by the data link layer, such as error handling and medium access control.	- Lectures	- Exams - Quizzes/itbls	60% of student achieve 60% of the CLO	Digital literacy

Teaching and Learning Resources:

Textbook(s):

Ross, Keith W., and James F. Kurose. (2021). *Computer networking A Top-Down Approach* (8th Edition). Addison-Wesley. ISBN: 978-1292405469

Teaching and Learning:

Blackboard Collaborate Ultra will be used to deliver the lectures, in addition to face-to-face lectures. The recordings of the lectures will be made available to the students automatically through the blackboard. All course material will be distributed online using the blackboard. Mid Term and Final Exams will follow the University's regulations with regards to the online or in-person exams. All submissions from the students for the in-class assessment, or homework assignments will be online

using the blackboard. Students will be provided with as many online resources as possible to facilitate their learning.

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. They can work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Academic Integrity, Collaboration:

All students are expected to maintain the utmost level of academic integrity. Any attempt to cheat will lead to appropriate sanctions. Substantial collaboration, in particular, in writing up solutions or during programming assignment, is not appropriate, nor is of course copying classmates' solutions. It is also never appropriate to search for solutions online and try to pass them off as your own work.

Students may discuss problems with their fellow students and the course staff, but they must write their own solutions. They may also consult any publications, but clearly cite any collaborators or outside sources.

In-class Activities/Teaching Methodology:

Students are expected to do their own work individually and/or in teams that will be graded. Activities include quizzes or in-class team-based learning (itbls), individual assignments, and team assignments.